

Step 1: the growth exponent α

from recurrent churn to hegemonic lock-in

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Stepwise-progression note, June 2026

Folder *stepwise-progression/*. Code: *step1.alpha.py*. Builds on *ANALYSIS_step0.tex*.

Abstract

We generalise the growth drift to $dS_i = \gamma_i R S_i^\alpha dt$ (+ demographic noise) and ask how the exponent α controls the rise and fall of dominant states. The answer is governed by the *relative* dynamics: the log-ratio of any two states has a growth-drift $\propto S_i^{\alpha-1} - S_j^{\alpha-1}$, whose sign flips at $\alpha = 1$. For $\alpha < 1$ a lead is *eroded* (sub-linear returns let laggards catch up $\Rightarrow$ recurrent turnover, the Step-0 churn); at $\alpha = 1$ the log-ratio is driftless (marginal); for $\alpha > 1$ a lead is *amplified* (super-linear returns $\Rightarrow$ a durable hegemon). Simulations confirm the mean leader lifetime rising from ~ 4 at $\alpha = 0$ to ~ 100 (half the run) at $\alpha = 2$, with the transition at $\alpha = 1$. A second, distinct threshold $\alpha = \frac{1}{2}$ (drift vs. noise of a single state) governs absolute, not relative, stability.

1 Model

$$R = \max\left(1 - \frac{1}{K} \sum_j S_j, 0\right), \quad dS_i = \gamma_i R S_i^\alpha dt + \sigma \sqrt{S_i} dW_i, \quad (1)$$

with γ_i a positive-mean OU process (so growth is sustained and α is isolated). $\alpha = 0$ recovers the additive Step-0 model; $\alpha = 1$ is multiplicative growth. We keep the demographic noise $\sigma \sqrt{S_i}$ fixed and vary only α .

2 Phenomenology

The character of the dynamics changes smoothly with α (Fig. 1, 20 seeds):

α	0	0.25	0.5	0.75	1.0	1.25	1.5	2.0
leadership changes	67	50	50	54	47	33	33	14
mean leader lifetime	3.8	5.3	6.4	8.1	10.9	21.9	41.8	99.2

Inequality is high (Gini ≈ 0.85 , top share ≈ 0.85) at *all* α : condensation onto a few states is the generic Step-0 effect of the $\sqrt{S}$ noise. What α tunes is not the *degree* of concentration but its *persistence* — whether the condensate moves (churn) or sticks (lock-in).

3 The relative dynamics: critical $\alpha = 1$

Leadership is a question about *ratios* of strengths, so we study the log-strength. By Itô on (1) with $f = \ln S$ ($f' = 1/S$, $f'' = -1/S^2$, $(dS)^2 = \sigma^2 S dt$),

$$d \ln S_i = \left[\gamma_i R S_i^{\alpha-1} - \frac{\sigma^2}{2S_i} \right] dt + \sigma S_i^{-1/2} dW_i. \quad (2)$$

Step 1: growth exponent α tunes recurrent churn $\rightarrow$ hegemonic lock-in (critical $\alpha = 1$)

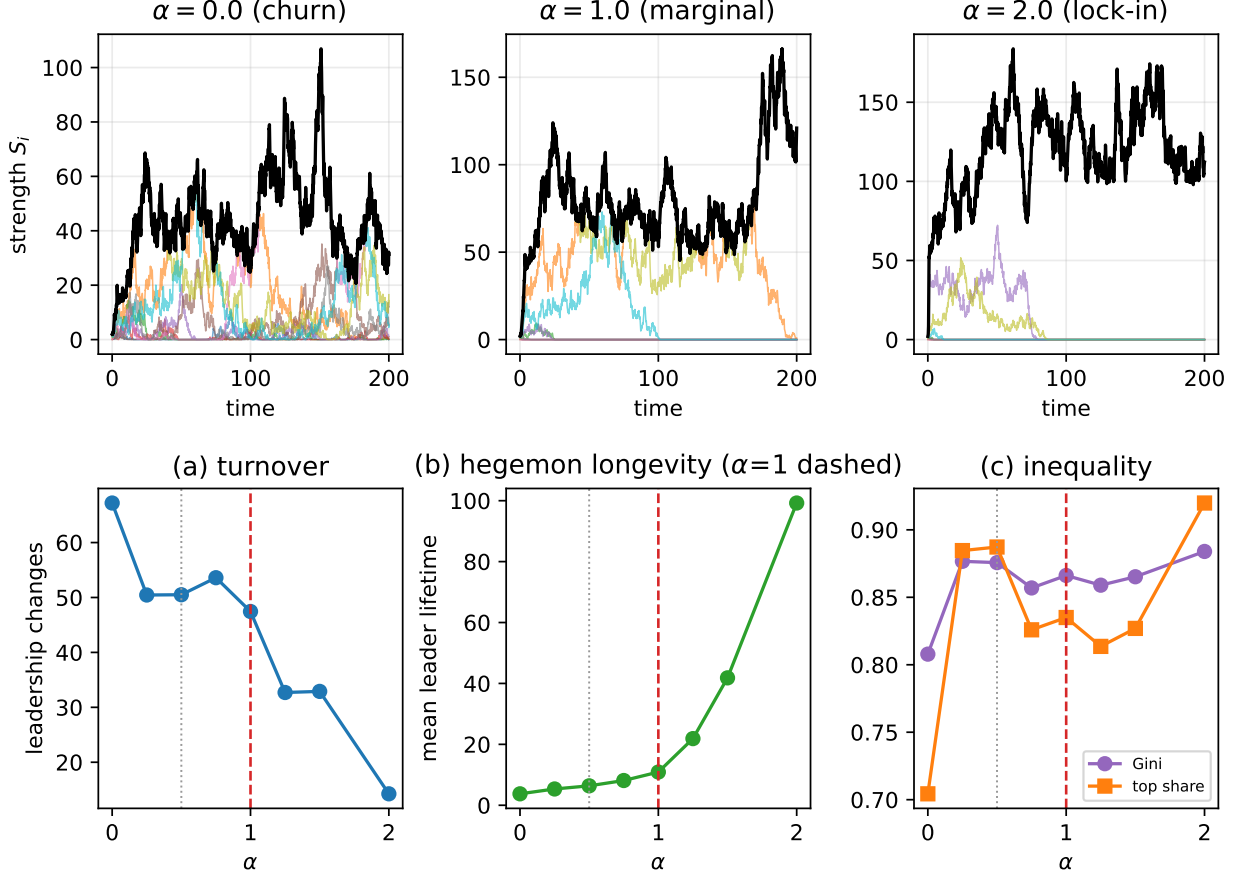


Figure 1: Top: example trajectories (leader in black) at $\alpha = 0$ (churn), 1 (marginal), 2 (lock-in). Bottom: leadership changes (a), mean leader lifetime (b), and inequality (c) vs α ; the dashed red line marks the relative-lock-in transition $\alpha = 1$, the dotted grey line the absolute-stability threshold $\alpha = \frac{1}{2}$.

For two states with a common rate γ , the log-ratio $u = \ln(S_i/S_j)$ obeys

$$du = \underbrace{\gamma R(S_i^{\alpha-1} - S_j^{\alpha-1})}_{\text{growth}} dt - \underbrace{\frac{\sigma^2}{2} \left(\frac{1}{S_i} - \frac{1}{S_j} \right)}_{\text{Itô (subdominant)}} dt + \sigma(S_i^{-1/2} dW_i - S_j^{-1/2} dW_j). \quad (3)$$

The leading (growth) drift contains $g(S) = S^{\alpha-1}$, with $g'(S) = (\alpha-1)S^{\alpha-2}$. Hence, taking $S_i > S_j$ (state i ahead):

$$\text{sign}(\text{growth drift of } u) = \text{sign}(\alpha - 1). \quad (4)$$

- $\boxed{\alpha < 1}$ — *diminishing returns*: g decreasing, so the leader's per-capita growth $\gamma R S^{\alpha-1}$ is *smaller* than the laggard's. The log-ratio mean-reverts toward 0: leads erode, the laggard catches up, and the demographic noise then reshuffles the order — **recurrent churn** (this includes Step 0, $\alpha = 0$).

- $\boxed{\alpha = 1}$ — *constant returns*: $g \equiv 1$, the growth drift of u vanishes and u is a (noise-driven) martingale — **marginal**. Turnover still occurs but only by diffusion of the ratio, so reigns lengthen.
- $\boxed{\alpha > 1}$ — *increasing returns*: g increasing, the leader grows faster *per capita*; the log-ratio drift is positive and the lead is amplified — **hegemonic lock-in**.

The Itô term $-\frac{\sigma^2}{2}(S_i^{-1} - S_j^{-1}) > 0$ for $S_i > S_j$ adds a small leader-stabilising drift that vanishes as the states grow; it shifts the balance slightly but does not change the $\alpha = 1$ sign flip. The noise variance $\sigma^2(S_i^{-1} + S_j^{-1})$ is dominated by the *smaller* state, so relative fluctuations are largest among small states — turnover happens while the contenders are still small.

Equivalent deterministic picture (a bifurcation at $\alpha = 1$). Drop the noise and let $\rho = S_i/S_j$. From (1),

$$\frac{d}{dt} \ln \rho = \gamma R (S_i^{\alpha-1} - S_j^{\alpha-1}) = \gamma R S_j^{\alpha-1} (\rho^{\alpha-1} - 1). \quad (5)$$

The fixed point $\rho = 1$ (equality) is *stable* for $\alpha < 1$ and *unstable* for $\alpha > 1$: a pitchfork-like exchange of stability in relative position. For $\alpha > 1$ any initial lead runs away ($\rho \rightarrow \infty$); indeed the bare ODE $\dot{S} = \gamma R S^\alpha$ has super-linear, finite-time-blow-up character, capped here only by $R \downarrow 0$ — the hegemon saturates at the resource limit rather than diverging.

4 The other threshold: $\alpha = \frac{1}{2}$ (absolute stability)

A *single* state’s drift-to-noise ratio is

$$\frac{\text{drift}}{\text{noise rate}} = \frac{\gamma R S^\alpha}{\sigma \sqrt{S}} = \frac{\gamma R}{\sigma} S^{\alpha-1/2}, \quad (6)$$

which is scale-invariant at $\alpha = \frac{1}{2}$. For $\alpha < \frac{1}{2}$ a large state is noise-dominated (its *absolute* size is hard to maintain — relevant to whether a state can climb from small values or is pushed toward 0); for $\alpha > \frac{1}{2}$ size is self-reinforcing. This is a statement about absolute sizes, *not* about who leads, and it is why the visible turnover transition sits at $\alpha = 1$ (relative) while inequality is already high well below it. Step 0’s collapse argument ($\alpha = 0 < \frac{1}{2}$, a driftless $\sqrt{S}$ Bessel martingale absorbed at 0) is the extreme case.

5 How α changes the rise and collapse of nations

- $\alpha < 1$ (*diminishing returns*). Leads erode; dominance is short and recurrent. Many states take turns on top; collapses are frequent and shallow. Mean reign $\sim$ a few time units. The natural “balance-of-power” regime.
- $\alpha = 1$ (*constant returns*). Marginal: the relative order is a martingale, so turnover is slow (reigns $\sim$ tens of units) and driven purely by fluctuations — no mechanism either erodes or entrenches a lead.
- $\alpha > 1$ (*increasing returns*). A single hegemon locks in and persists for most of the observation window; collapses become *rare* but, when a large fluctuation finally unseats the giant, *deep* — long, dramatic hegemonic cycles rather than rapid churn.

Empirically (Fig. 1b) the mean reign grows by $\sim 25\times$ across $0 \leq \alpha \leq 2$, with the knee at $\alpha = 1$, exactly as §3 predicts.

6 Caveat and outlook

The $\max(\mathbf{S}, 0)$ clamp again injects a little mass under the $\sqrt{S}$ noise (see Step 0); use a positivity-preserving CIR scheme for long-time quantitative work. The exponent α is the cleanest single knob for the rise-and-fall character and is a natural control parameter to carry into the later steps: a *conflict* mechanism (Step 2) that transfers strength to the winner is itself an effective increasing-returns ($\alpha > 1$ -like) force, so we expect war to push an otherwise churning ($\alpha < 1$) world toward lock-in — a competition between the growth exponent and the conflict rule that the next step will make explicit.